Prepared for submission to JHEP

November 8, 2026
This version: November 8, 2026

Light vectors and scalar interactions near heterotic K3 gauge enhancement

Bernd Johannes Wuebben

New York, NY, USA

E-mail: wuebben@gmail.com

Abstract: We determine light-vector masses and a leading neutral-scalar interaction near gauge enhancement in an $SO(32)$ heterotic compactification on a smooth K3 surface. A family of stable bundles approaches a repeated rank-three factor. For harmonic deformation amplitudes a, b, quantitative Hermitian–Einstein reconstruction gives three leading squared vector masses proportional to b^2, a^2 and a^2+b^2, with relative geometric error of order a^2+b^2 independent of b/a. We calculate and bound the first tree-level correction to the effective vector eigenvalue equation, including the complete leading metric and two-form Kaluza–Klein towers. An anti-self-dual Hodge identity gives positivity and preserves the unbroken gauge kernels. In the same family, a neutral-scalar interaction coefficient grows as the inverse square of the deformation amplitude at fixed positive b/a, with quadratic relative geometric error. Its identification uses derivative estimates for the bundle metric and a reduction retaining the other neutral fields. The vector bound applies to the stated first-order system. For the scalar result, we first extract the tree-level large-volume coefficient at each regular point, then approach enhancement at fixed positive ratio; a uniform finite-string-scale remainder is not asserted.

Keywords: Superstrings and Heterotic Strings, Compactification and String Models, Supersymmetric Effective Theories

Contents

1 Introduction **1**

2 The compactification and its enhancement locus **3**
2.1 Stable factors and the orthogonal embedding 3
2.2 Harmonic coordinates and actual bundles 5

3 Light vectors and the retained spectrum **7**
3.1 The physical vector operator 7
3.2 Massless fields and Higgs multiplets 8
3.3 The common low-energy space 9

4 The first coupled vector correction **9**
4.1 Action convention and the corrected background 9
4.2 The two heavy spin-one sectors 11
4.3 Hodge cancellation, positivity and the error bound 12

5 Derivative estimates and a nonzero curvature component **13**
5.1 Preserving the quadratic rate in holomorphic coordinates 13
5.2 What fails near a changing stabilizer 15
5.3 Curvature in additional harmonic deformation directions 15

6 A leading neutral-scalar interaction **17**
6.1 Partial reduction on a regular patch 17
6.2 Mixed constraints and the vertical kinetic coefficient 17
6.3 The other neutral fields and ambient curvature 19
6.4 The interaction and its order of limits 20

7 Discussion **21**

1 Introduction

Gauge enhancement in a heterotic compactification occurs when the internal connection acquires additional parallel adjoint sections. A deformation away from that locus makes the corresponding external vectors massive. The finite-dimensional Higgs mass matrix describes the leading mechanism, but an absolute error estimate need not resolve its smallest eigenvalue: several Higgs scales can approach zero at unrelated rates. There is a related problem

for scalar interactions. A comparison of kinetic metrics does not, by itself, control the derivatives entering their curvature or identify the curvature of a bundle-moduli subspace with a coupling in the full reduced theory.

We address these questions on one family of smooth K3 compactifications of the $\mathrm{Spin}(32)/\mathbb{Z}_2$ heterotic theory with vector structure. We use SO(32) for its gauge algebra and its real vector-bundle description. The background bundle contains two identical stable rank-three factors at the enhancement point. Two orthonormal harmonic deformation coordinates, with coefficients a, b, break the multiplicity SU(2). The K3 surface remains smooth and its polarization is fixed. The moment maps vanish throughout this bundle enhancement; no nonzero Fayet–Iliopoulos parameter is varied.

Let g_0 be the fixed reference K3 metric, $V_0 = \mathrm{Vol}(g_0)$, and let $L^2 g_0$ be the physical internal metric at leading order. The three leading squared vector masses obey

$$m_j^{2,(0)} = \widehat{m}_j^2\big(1 + O(a^2 + b^2)\big), \qquad (\widehat{m}_1^2, \widehat{m}_2^2, \widehat{m}_3^2) = \frac{8}{3V_0 L^2}(b^2, a^2, a^2 + b^2), \quad 0 < b \le a. \tag{1.1}$$

The error is proportional to each individual squared mass, with a constant independent of b/a. In particular, at $a = \varepsilon$, $b = \varepsilon^p$, $p > 1$, the weakest squared mass has order ε^{2p}/L^2, while its relative geometric error still has order ε^2. This is a statement about actual Hermitian–Einstein connections, obtained by applying the quantitative local comparison of [1] to the compactification family. The Pauli-matrix eigenvalues alone would not give this conclusion.

We also calculate the first coefficient of the coupled tree-level vector equations. Write $\eta = \alpha'/L^2$. Eliminating the complete leading metric and two-form vector towers gives a finite, energy-dependent pencil $F^{[1]}(\lambda, \eta)$, defined in section 4. Its three positive roots, converted to squared masses by L^{-2}, satisfy

$$\left| \frac{m_j^{2,[1]}}{\widehat{m}_j^2} - 1 \right| \le C\left(a^2 + b^2 + \frac{\eta}{b^2}\right). \tag{1.2}$$

An anti-self-dual (ASD) Hodge identity checks the relative coefficients of the two mixings and proves positivity and preservation of the unbroken kernels. The sufficient region $\eta = o((a^2 + b^2)b^2)$ makes this calculated correction smaller than the geometric relative error. Equation (1.2) concerns the defined pencil; it does not bound an uncomputed remainder in exact string poles.

The second physical quantity is a neutral-scalar interaction. In the same holomorphic local quotient chart, the actual harmonic metric has a relative quadratic comparison with the hyperkähler cone metric. Ricci-flat Kähler regularity upgrades this comparison to every fixed derivative order on compact smooth angular regions. An explicit horizontal two-plane has cone sectional curvature $3/(2b^2)$. We solve the mixed Einstein and two-form constraints, account for the other neutral fields and for the small axion kinetic block, and obtain the leading physical sectional curvature

$$\mathcal{K}_{\mathrm{phys}}^{\mathrm{lead}} = \frac{3g_{10}^2}{4L^2\kappa^2\varepsilon^2}\big(1 + O_\kappa(\varepsilon^2)\big), \qquad a = \varepsilon, \quad b = \kappa\varepsilon, \quad 0 < \kappa \le 1. \tag{1.3}$$

Here g_{10} is defined by the gauge action in section 3. This coefficient multiplies a specified two-polarization quartic interaction in scalar normal coordinates. Its derivation first extracts the tree-level large-volume coefficient on a fixed regular patch, then takes the displayed geometric limit. It is not uniform as $\kappa \to 0$. Eliminating the Higgs multiplets requires probe energies much smaller than b/L.

Heterotic bundle stability, charged fields and D-term Higgsing have an established effective-theory description [2]. The SO(32) embedding, spectrum and abelian mass criteria used here follow the K3 analysis of [3]. Bundle and matter metrics were computed in [4], while the general geometry–bundle–two-form fibration and its large-volume hierarchy were developed in [5]. We retain the compensator and curvature-squared terms needed to identify the leading coefficient; we do not import a specialized duality completion as the metric of the present compactification. The relation between scalar curvature and covariant amplitudes is also standard [6].

Universal heterotic representatives and higher-order moduli metrics are developed in [7–9]. Their Calabi–Yau threefold results concern four-dimensional compactifications; we use the action and covariant-variation conventions, not their moduli-metric theorems as results for six-dimensional K3 vacua.

The quantitative input is the energy-relative spectral and quotient-metric comparison in [1]. We use the local harmonic quadraticity argument of [10] without extending that paper's global compactness theorem to this family. The new application combines those inputs with a compact heterotic model, the coupled vector calculation, and a physically normalized curvature coefficient on that same family. The Hodge identity, interior estimates and quotient curvature formula are standard mechanisms or direct calculations. Their role is to make the physical identifications and their accuracy precise.

Section 2 constructs the family and fixes its local coordinates. Sections 3 and 4 derive the vector statements and the retained field content. Sections 5 and 6 establish the derivative estimates and the leading ambient scalar interaction. Throughout, the string coupling is taken in the tree regime independently of the geometric parameters.

2 The compactification and its enhancement locus

2.1 Stable factors and the orthogonal embedding

Fix a smooth projective K3 surface S with $\mathrm{Pic}(S) = \mathbb{Z}H$, where H is ample. Let g_0 be its Ricci-flat Kähler metric in that class, oriented so that primitive $(1,1)$ forms are anti-self-dual. Norms and harmonic representatives without an explicit scale use g_0 and unnormalized trace inner products. Write $V_0 = \mathrm{Vol}(S, g_0)$.

Choose locally free μ_H-stable bundles A, B with

$$\begin{aligned} \operatorname{rk} A = \operatorname{rk} B = 3, \qquad & \det A = \det B = \mathcal{O}_S, \\ c_2(A) = 7, \quad c_2(B) = 10, \qquad & A \not\simeq A^*, \quad B \not\simeq B^*. \end{aligned} \tag{2.1}$$

Here existence includes the locally free, slope-stable and non-self-dual conditions. We give the argument because these conditions determine both the analytic application and the unbroken gauge algebra.

For $k = 7, 10$ the Mukai vector is $v_k = (3, 0, 3-k)$, with $v_7^2 = 24$ and $v_{10}^2 = 42$. Both vectors are primitive. An equal reduced-Hilbert-polynomial proper subsheaf would have first Chern class zero and Euler characteristic divided by rank equal to $-1/3$ or $-4/3$. Neither is possible for rank one or two. Thus H is general for these vectors. Yoshioka's nonemptiness and irreducibility theorem, together with the slope-stability and local-freeness criteria in [11, Theorem 0.1, Proposition 0.5 and section 3.3], gives nonempty slope-stable and locally free opens. In that notation the integer l equals three and $v_k^2 \geq 2l^2$. The exceptions to the locally free criterion do not occur: the vectors have rank three, whereas the spherical exception of the form $(\mathrm{rk}\, v_0)v_0 - \omega$ would have square rank. Irreducibility makes the two opens intersect. Since $\mathrm{Pic}^0(S) = 0$, their determinant is trivial.

Dualization acts holomorphically as an involution on this smooth open. At a fixed point F, simplicity makes an isomorphism $F \to F^*$ symmetric or skew-symmetric after scalar normalization. Odd rank excludes the latter. The invariant tangent space is $H^1(S, \mathfrak{so}(F)) \simeq H^1(S, F)$: use $\Lambda^2 F^* \simeq F$ and the symmetric form. Stability and Serre duality give $H^0(F) = H^2(F) = 0$, and Riemann–Roch gives

$$h^1(F) = k - 6, \qquad h^1(\mathrm{End}\, F) = 6k - 16. \tag{2.2}$$

The fixed locus of a holomorphic involution is smooth with its invariant tangent space. One can see this directly by replacing local coordinates z centered at a fixed point by $\frac{1}{2}(z + D\iota_0\, z \circ \iota)$; their derivative is the identity and the action becomes linear. The fixed locus therefore cannot fill the stable locally free open. This proves that the choices in (2.1) exist. Their distinct second Chern classes also exclude isomorphisms between A or A^* and B or B^*.

At enhancement set $E_0 = A \otimes \mathbb{C}^2$ and define the real gauge bundle

$$\mathcal{V} = (E_0 \oplus B)_{\mathbb{R}} \oplus \mathbb{R}^{14}, \qquad \mathcal{V} \otimes \mathbb{C} = E_0 \oplus E_0^* \oplus B \oplus B^* \oplus \mathcal{O}^{14}. \tag{2.3}$$

It has real rank 32, $w_2(\mathcal{V}) = 0$, and $p_1(\mathcal{V}) = -2c_2(E_0 \oplus B) = -48 = p_1(TS)$. The weighted instanton number is

$$2c_2(A) + c_2(B) = 2 \cdot 7 + 10 = 24. \tag{2.4}$$

There is consequently no five-brane source in the integrated Bianchi identity [3]. The Donaldson–Uhlenbeck–Yau correspondence [12, 13] supplies determinant-flat HE connections, which are ASD on S. They give the leading background with eight supercharges, or six-dimensional $\mathcal{N} = (1, 0)$ supersymmetry. Equation (2.4) is an integrated condition; it does not set the gauge and tangent curvature densities equal pointwise. Their difference sources the first correction in section 4.

The parallel orthogonal adjoint sections have compact algebra

$$\mathfrak{h}_0 = \mathfrak{so}(14) \oplus \mathfrak{u}(2) \oplus \mathfrak{u}(1), \qquad \dim_{\mathbb{R}} \mathfrak{h}_0 = 96. \tag{2.5}$$

Indeed the endomorphism blocks give $\mathrm{End}(\mathbb{C}^2)$ and $\mathbb{C}$ by simplicity. The additional exterior-square blocks have no parallel sections. For example,

$$\Lambda^2(A\otimes\mathbb{C}^2)=(\Lambda^2A\otimes\mathrm{Sym}^2\,\mathbb{C}^2)\oplus(\mathrm{Sym}^2\,A\otimes\Lambda^2\mathbb{C}^2),$$

and $H^0(A\otimes A)=\mathrm{Hom}(A^*,A)=0$ by stability and non-self-duality. The other cross blocks vanish for the same reason. There are no maps to the trivial fourteen-dimensional factor. This also explains why replacing A by a pseudoreal rank-two factor would give a different centralizer. All central factors here have $c_1=0$; the abelian flux mass matrix of [3, section 3.2] vanishes. Protection of the nonconstant-profile vector calculation requires the separate argument in section 4.

2.2 Harmonic coordinates and actual bundles

General K3 formality results [14] identify quadratic deformation germs. The quantitative metric application below needs the stronger coordinate specification in the original harmonic chart.

The harmonic space $H^1(S,\mathrm{End}\,A)$ has complex dimension 26 and a quaternionic structure. Choose thirteen orthonormal symplectic pairs, normalized using the parallel holomorphic two-form. Then

$$H^1(S,\mathrm{End}\,E_0)=H^1(S,\mathrm{End}\,A)\otimes\mathrm{End}(\mathbb{C}^2)$$

has coordinates $(X_j,Y_j)_{j=1}^{13}$ with the flat real metric

$$\|\dot x\|^2=\mathrm{Re}\sum_{j=1}^{13}\mathrm{tr}(\dot X_j^*\dot X_j+\dot Y_j^*\dot Y_j). \tag{2.6}$$

The multiplicity group acts by simultaneous conjugation; its center acts trivially. The complex and real moment equations are

$$\sum_j[X_j,Y_j]=0,\qquad \sum_j\big([X_j,X_j^*]+[Y_j,Y_j^*]\big)=0. \tag{2.7}$$

Only their zero level is used. Let $\mathcal{C}$ be this hyperkähler quotient by the effective group $\mathrm{PU}(2)$, with metric $g_{\mathcal{C}}$ on its smooth-simple part.

We record why these are exact local deformation coordinates, rather than merely the quadratic part of an unspecified obstruction. Put $d=\bar\partial_{E_0}$ and denote harmonic projection and Green operators by H_q,G_q. The convergent harmonic recursion of [1, Lemma 2.1] is

$$\mathfrak{a}(x)=x-d^*G_2(\mathfrak{a}(x)\wedge\mathfrak{a}(x)),\qquad \mathfrak{k}(x)=H_2(\mathfrak{a}(x)\wedge\mathfrak{a}(x)). \tag{2.8}$$

On a Ricci-flat K3 its obstruction is exactly

$$\mathfrak{k}(x)=H_2(x\wedge x). \tag{2.9}$$

For completeness, a harmonic $\operatorname{End} E_0$-valued $(0,2)$-form is $\xi\bar{\sigma}$ with ξ parallel. This follows from the formula for $d^*(\xi\bar{\sigma})$ and the Bochner identity on the zero-mean-curvature endomorphism bundle. If $d^*u = 0$ and $\gamma = \phi\bar{\sigma}$, the coefficients of $u \wedge d^*\gamma$ and $d^*\gamma \wedge u$ are, up to common nonzero scalars, $-\sum_j u_{\bar{j}} \nabla_j \phi$ and $\sum_j (\nabla_j \phi) u_{\bar{j}}$. Pair either with a parallel ξ and integrate by parts, keeping the matrix order. The resulting terms contain $\sum_j \nabla_j u_{\bar{j}} = -d^*u$ and vanish. Consequently both products have zero harmonic projection. Writing $\mathfrak{a} = x + \beta$ with β coexact and d^*-closed eliminates all terms containing β in the obstruction and proves (2.9). This is precisely the local argument in [10, Lemma 3.1]; it uses none of that paper's later global rank or no-bubbling hypotheses.

Serre duality identifies (2.9) with the first equation in (2.7); a common nonzero symplectic factor and the rank-dual identification do not change its zero set. Fixing the determinant removes no harmonic directions because $H^1(S, \mathcal{O}) = 0$. The analytic quotient construction [15] and HE reconstruction [1, Theorem 1.1] therefore identify the balanced quotient with the actual nearby HE moduli. This identification Ψ is holomorphic in these same coordinates: the holomorphic classifying map and the reconstructed HE map assign the same bundle class to each harmonic deformation. No further nonlinear reparametrization is made.

The quantitative input, specialized to this fixed background, is

$$(1 - Cr^2) g_{\mathcal{C}} \;\le\; \Psi^* g^{\mathrm{KS}} \;\le\; (1 + Cr^2) g_{\mathcal{C}}, \qquad r = \|x\|, \tag{2.10}$$

on genuine smooth stratum tangents. Here g^{KS} is the real harmonic Kodaira–Spencer metric in the unnormalized Dolbeault norm. The same constant works when positive orbit scales have arbitrary ratios. It may depend on the fixed metric, bundles, ranks, harmonic dimensions and elliptic gaps. This does not assert uniformly sized smooth coordinate charts through every lower stratum. On the smooth-simple locus g^{KS} is hyperkähler [16].

Let σ_j be the Pauli matrices with $\operatorname{tr}(\sigma_i \sigma_j) = 2\delta_{ij}$ and $[\sigma_i, \sigma_j] = 2i\epsilon_{ijk}\sigma_k$. Our family is

$$X_1 = a\sigma_3, \qquad X_2 = b\sigma_1, \qquad X_j = 0 \ (j \ge 3), \qquad Y_j = 0, \qquad a, b \in \mathbb{R}. \tag{2.11}$$

It satisfies both moment equations since its nonzero matrices are Hermitian. If $ab \neq 0$, a line invariant under σ_3 is an eigenline, and σ_1 exchanges the two eigenlines. The representation is therefore simple. Reconstruction gives an actual stable rank-six bundle $E_{a,b}$ with $c_2 = 14$ and trivial determinant. The local simple moduli space has complex dimension $104 - 2 \cdot 3 = 98$.

On a nonzero axis the representation is the sum of two distinct characters, corresponding to two nearby stable rank-three factors. Upper semicontinuity preserves the other Hom vanishings after shrinking the neighborhood. The unbroken algebra is thus $\mathfrak{so}(14) \oplus \mathfrak{u}(1)^2$ generically, $\mathfrak{so}(14) \oplus \mathfrak{u}(1)^3$ on an axis, and (2.5) at the origin. Their dimensions are 93, 94 and 96, respectively.

3 Light vectors and the retained spectrum

3.1 The physical vector operator

On real skew-adjoint generators set $B_F(U,V) = -\frac{1}{2}\operatorname{Tr}_{32}(UV)$. Define g_{10} by

$$S_{\mathrm{YM}} = -\frac{1}{4g_{10}^2}\int B_F(F_{MN}, F^{MN})\, d\operatorname{Vol}_{10}\,. \tag{3.1}$$

The constant dilaton is absorbed in this coupling. Initially use the product spacetime metric $g_{1,5} + L^2 g_0$; a constant change to six-dimensional Einstein frame must rescale masses and scalar kinetic tensors consistently. On a realified unitary block B_F restricts to $-\operatorname{Re}\operatorname{Tr}$. A multiplicity generator at E_0 consequently has reference norm $3V_0 \operatorname{tr}_2(\xi^*\xi)$.

For a Lorentz-transverse external fluctuation $\delta\mathcal{A}_\mu = v_\mu\psi(y)$, background gauge separates the internal Goldstone variation. The external kinetic and mass forms are $g_{10}^{-2}\|\psi\|^2_{L^2(L^2 g_0)}$ and $g_{10}^{-2}\|d_{\mathcal{A}}\psi\|^2_{L^2(L^2 g_0)}$. The coupling cancels in their generalized eigenvalue. On endomorphism zero-forms the HE Kähler identity gives

$$d^*_{\mathcal{A}} d_{\mathcal{A}} = 2\Delta_{\bar\partial,\operatorname{End} E}. \tag{3.2}$$

The induced endomorphism mean curvature is zero, so the ∂ and $\bar\partial$ Laplacians agree in this degree. This is a zero-form vector-profile identity, not an identity for the full one-form Yang–Mills Hessian; compare [4, appendix A]. The zero-form and one-form norms scale as L^4 and L^2, respectively, so physical squared masses acquire the factor L^{-2}.

For $\xi = \sum z_j\sigma_j$ the finite Dolbeault energy and mass are

$$q_x(\xi) = \sum_j \|[\xi, X_j]\|^2 + \|[\xi, Y_j]\|^2 = 8\big(a^2|z_1|^2 + (a^2+b^2)|z_2|^2 + b^2|z_3|^2\big), \tag{3.3}$$

$$M_0(\xi) = 3V_0 \operatorname{tr}(\xi^*\xi) = 6V_0 \sum_j |z_j|^2. \tag{3.4}$$

Thus its ordered eigenvalues for $0 < b \le a$ are $\frac{4}{3V_0}(b^2, a^2, a^2+b^2)$.

Proposition 3.1. *For the actual bundles of (2.11), sufficiently near the origin, there are exactly three real vector species whose positive masses tend to zero away from the unbroken kernel. They satisfy (1.1), with a constant independent of b/a. The corresponding two-sided inequality retains exact zeros on the axes and at the origin. All other nonzero vector-profile eigenvalues have a fixed positive lower bound in reference units.*

Proof. Apply [1, Theorem 1.1(ii)] at the locally free polystable bundle $A \otimes \mathbb{C}^2$, at constant polarization and fixed determinant. Its trace-free multiplicity algebra has dimension three. The theorem bounds every small Dolbeault eigenvalue between $(1 - C\|x\|^2)\nu_j(x)$ and $(1 + C\|x\|^2)\nu_j(x)$, including zeros, and gives a positive complementary gap. Here $\|x\|^2 = 2(a^2+b^2)$ and (3.3)–(3.4) give ν_j. Equation (3.2) and the internal scale give (1.1). The real adjoint sector is preserved by this Laplacian. Each complex eigenspace is the complexification of its

Table 1. Generic massless hypermultiplet contributions. Charges refer to the two central U(1) factors. The fourteen-dimensional sectors contain hypermultiplets in the vector representation of SO(14), not additional gauge vectors.

Sector	Dimension-weighted hypermultiplet count
E moduli, neutral	49
B moduli, neutral	22
$\Lambda^2 E$, charge $(2,0)$	26
$\Lambda^2 B$, charge $(0,2)$	4
$E \otimes B$, charge $(1,1)$	66
$E \otimes B^*$, charge $(1,-1)$	66
E sector in **14**, charge $(1,0)$	28
B sector in **14**, charge $(0,1)$	56
K3 geometry and two-form fields	20
Total	337

Hermitian, or equivalently skew-Hermitian, real form. Its complex multiplicity is therefore the number of real generator profiles, not twice that number. The remaining orthogonal-adjoint blocks have the kernels already identified in (2.5); their positive spectrum stays separated from zero by elliptic continuity on the fixed compact background. □

The essential input is the relative form of the imported estimate. In its proof the complementary Schur term is bounded by the finite energy of the same vector. A uniform additive error of order $(a^2+b^2)^2$ would not establish the result when $b \ll a^2$. There is no additional gauge coupling multiplying the masses in (1.1) after the convention (3.1) is fixed.

3.2 Massless fields and Higgs multiplets

The vector gap does not make the remaining matter heavy. The full orthogonal adjoint decomposes into endomorphism, exterior-square, mixed tensor and fourteen-dimensional vector sectors. For a stable determinant-trivial rank-r bundle F with $c_2(F) = k$ and the Hom vanishings established above, Riemann–Roch and Serre duality give

$$\begin{aligned} h^1(F) &= k - 2r, & \dim_{\mathbb{H}} \mathcal{M}_F &= rk - r^2 + 1, \\ h^1(F \otimes G) &= r_G k_F + r_F k_G - 2r_F r_G, & h^1(\Lambda^2 F) &= (r-2)k - r(r-1). \end{aligned} \tag{3.5}$$

These follow by integrating $\mathrm{ch}(\cdot)\,\mathrm{td}(S)$; the exterior-square formula uses $c_2(\Lambda^2 F) = (r-2)c_2(F)$ when $c_1(F) = 0$. The neutral moduli formula follows from $h^1(\mathrm{End}\, F) = 2rk - 2r^2 + 2$. In the spectrum convention of [3], complex conjugate sectors combine into full hypermultiplets. At a generic point the dimension-weighted counts are listed in table 1.

At enhancement the E endomorphism sector consists of thirteen adjoint-U(2) hypermultiplets and contributes 52. On a nonzero axis the two distinct rank-three factors contribute $2 \cdot 13$

neutral hypermultiplets and 24 off-diagonal charged ones, giving 50. The other dimension-weighted totals are unchanged. Thus

$$(n_H, n_V) = \begin{cases} (340, 96), & a = b = 0, \\ (338, 94), & ab = 0, \quad a^2 + b^2 > 0, \\ (337, 93), & ab \neq 0. \end{cases} \tag{3.6}$$

There is one tensor multiplet, containing the six-dimensional dilaton, and the gravity multiplet. Every row obeys $n_H - n_V + 29n_T = 273$. This is a gravitational anomaly count; it does not replace the gauge-anomaly and embedding analysis of [3]. As a separate index check, the real gauge ASD complex has $h^1 - h^0 - h^2 = 4 \cdot 30 \cdot 24 - 4 \cdot 496 = 896$ on K3. Our generic gauge hypermultiplets contribute $4(337 - 20)$, and $h^0 = 93$ gives $4(337 - 20) - 4 \cdot 93 = 896$; the corresponding enhanced and axis counts agree as well. Here the expression subtracts all four parallel contributions, equivalently using $h^2 = 3h^0$ in the ASD index.

Three hypermultiplets and three vectors combine into three massive $\mathcal{N} = (1, 0)$ vector multiplets generically. Each massive vector has five physical polarizations and three physical real scalar partners; the fourth scalar of its hypermultiplet is a Goldstone mode. On an axis only two such multiplets are massive.

3.3 The common low-energy space

Choose background gauge, diffeomorphism gauge and two-form gauge at the reference background. The leading internal operators are the adjoint zero-form Laplacian for vectors, the positive ASD deformation square for connection scalars, the physical Lichnerowicz operator for metric deformations, ordinary Hodge Laplacians for forms, and the corresponding Dirac complexes for fermions. The ASD square is $d_{\mathcal{A}} d_{\mathcal{A}}^* + 2 d_{\mathcal{A}}^{+*} d_{\mathcal{A}}^+$ on one-forms. The parallel-spinor factorization gives nonnegativity of the physical Ricci-flat metric operator [17]; this excludes the unphysical Euclidean conformal mode. Compactness gives finite kernels and positive complementary gaps. Supersymmetry identifies the physical superpartners.

Take a cutoff below the minimum of these reference gaps and retain every reference low mode, including the gauge/Goldstone complex. Smooth elliptic dependence gives common finite low spectral clusters and a complementary gap $\gamma > 0$ for all sufficiently small (a, b). One must not first divide out a different number of scalar gauge directions at each stratum and then assert continuity on a fixed space. The theory with the Higgs multiplets retained has the common leading window $E_{\text{probe}} \ll \sqrt{\gamma}/L$. Eliminating them instead requires $E_{\text{probe}} \ll b/L$ on a generic regular patch. Higher-derivative string terms are used perturbatively through leading resolvents, not as a positive second-order operator on the entire tower.

4 The first coupled vector correction

4.1 Action convention and the corrected background

For this calculation use unit-scale internal coordinates and remove the common gravitational prefactor. Set $\eta = \alpha'/L^2$ and use positive forms $B_F = -\frac{1}{2}\,\mathrm{Tr}_{32}$ and $B_R = -\frac{1}{2}\,\mathrm{Tr}_T$ on the

compact gauge and internal Euclidean rotation algebras. The Lorentzian extension of the tangent trace is understood in the ten-dimensional action. Our convention is

$$S = \int e^{-2\Phi}\sqrt{-G}\Big[R + 4|d\Phi|^2 - \frac{1}{12}H_{MNP}H^{MNP} - \frac{\eta}{4}\big(B_F(F_{MN}, F^{MN}) - B_R(R_{MN}, R^{MN})\big)\Big]. \tag{4.1}$$

Use the field choice in which no additional independent purely bosonic operator at order α'^2 is required beyond the recursively defined torsion and composite-connection terms [18, 19]. This statement does not discard their order-α'^2 contributions. The tangent connection is the composite Hull connection $\Theta_H = \Theta_{\mathrm{LC}} + \frac{1}{2}H$, where $H^a{}_b = H^a{}_{bM}dx^M$ in an orthonormal frame, in the convention of [9, section 2.1]. It reduces to Levi–Civita at leading order. Fix Chern–Simons signs by

$$dH = \frac{\eta}{2}\big(B_R(R\wedge R) - B_F(F\wedge F)\big), \tag{4.2}$$

$$\delta H = d\mathcal{B} - \eta B_F(\delta\mathcal{A}\wedge F) + \eta B_R(\delta\Theta\wedge R). \tag{4.3}$$

$\mathcal{B}$ denotes the gauge-covariantly shifted two-form variation. These choices fix relative normalizations rather than combining formulas with incompatible trace or dilaton conventions.

For $t = (a, b)$ let F_t be the full leading ASD curvature. Through first order the smooth background is

$$G_S = (1 + \eta f_t)g_0, \qquad \Phi = \Phi_0 + \tfrac{1}{2}\eta f_t, \qquad H = -\eta *_0 df_t, \tag{4.4}$$

where f_t has zero mean and solves

$$\nabla^m\nabla_m f_t = \frac{1}{4}\big(B_R(R_{mn}, R^{mn}) - B_F(F_{t,mn}, F_t^{mn})\big). \tag{4.5}$$

Here ∇^2 has the divergence sign; the nonnegative scalar Laplacian is $\Delta_0 = -\nabla^2$. The source has mean zero by (2.4). Scalar elliptic inversion gives a unique solution, with uniform spatial bounds on a sufficiently small family of reconstructed connections. These connections have uniformly bounded spatial coefficients in fixed gauge even though the small spectral ratios degenerate. Parameter derivative bounds used later are only on fixed regular patches. For sufficiently small η, $1 + \eta f_t > 0$.

The normalization can be checked directly. In four dimensions the linear conformal Ricci tensor is $-\nabla_m\nabla_n f - \frac{1}{2}g_{mn}\nabla^2 f$; the dilaton Hessian adds $\nabla_m\nabla_n f$. For an ASD curvature, $B_F(F_{mp}, F_n{}^p) = \frac{1}{4}g_{mn}B_F(F_{pq}, F^{pq})$. The Einstein equation from (4.1) gives (4.5). Also $B_F(F\wedge F) = -\frac{1}{2}B_F(F_{mn}, F^{mn})dV$ and $d(-*df) = -\nabla^2 f\, dV$, so the Bianchi identity gives the same equation. The dilaton equation gives the same trace condition, and the two-form equation holds because the leading weighted dual flux is exact. The gauge equation $D_{\mathcal{A}}(e^{-2\Phi} *_4 F) - e^{-2\Phi}F\wedge *_4 H = 0$ also cancels through this order, using $*_4 F = -F$ and $*_4 H = \eta df$. The corresponding conformal/dilaton ansatz is the usual first-order supersymmetric K3 construction [20]; ASD persists under conformal rescaling. It is a solution through the computed order, not an existence assertion for an exact finite-η background.

4.2 The two heavy spin-one sectors

Let P_t be the whole 96-dimensional low real spectral projection of $d^*_{\mathcal{A}_t} d_{\mathcal{A}_t}$ on the full orthogonal adjoint. Put $A_t = P_t d^*_{\mathcal{A}_t} d_{\mathcal{A}_t} P_t$. This space includes all persistent massless vectors. Define

$$T_t p = B_F(p, F_t) \in \Omega^2(S), \qquad (J_t p)_m = B_F(F_{t,mn}, D^n p), \qquad I_t v = \iota_v F_t. \tag{4.6}$$

The Yang–Mills equation and invariance of the trace give $J_t = d^* T_t = I_t^* d_{\mathcal{A}_t}$, and hence $d^* J_t = 0$. In components the latter follows by differentiating $B_F(F_{mn}, D^n p)$; the curvature commutator contracts to zero.

Set $s = \sqrt{\eta}$ and canonically normalize the gauge fluctuation by $p = s\,\delta\mathcal{A}_\mu$. Let v be the metric vector in the coframe $dy^m + v^m_\mu dx^\mu$, and let β be the shifted two-form vector. After the diffeomorphism and form constraints their profiles are co-closed real one-forms. At leading order both have ordinary L^2 kinetic norm and squared-mass operator $\Delta_1 = d^* d$ on co-closed one-forms. The external field-strength coefficient is $-1/4$ for each. For the two-form, the three placements of the internal index in $H_{\mu\nu m}$ turn its contribution to $-H^2/12$ into $-H_{\mu\nu m} H^{\mu\nu m}/4$, giving the external vector kinetic term. The three placements of the external index in $H_{\mu mn}$ instead give its mass term; the factor $1/2$ in the internal two-form norm yields the usual Proca normalization. For the metric the kinetic coefficient follows from the quadratic Kaluza–Klein curvature formula; on a Ricci-flat base its Killing-operator form reduces, on co-closed forms, to $\|dv\|^2$. K3 has no harmonic one-forms, and a Killing field on a compact Ricci-flat K3 would be parallel. Thus both sectors have a fixed positive gap.

With $I_t v = \iota_v F_t$ and the plus coframe just specified, the horizontal mixed gauge curvature is $-d_{\mathcal{A}} \delta\mathcal{A}_\mu - I_t v_\mu$. The terms determining the first low-vector coefficient are therefore

$$Q = \|d_{\mathcal{A}} p + s I_t v\|^2 + \|dv\|^2 + \|d\beta - s T_t p\|^2, \qquad K = \|p\|^2 + \|v\|^2 + \|\beta\|^2 \tag{4.7}$$

at leading heavy order. Reversing the coframe convention reverses the metric mixing sign and leaves its Schur term unchanged. Exact one-form gauge directions are not additional transverse vector species. Harmonic two-form components of $T_t p$, in contrast, must remain in the direct norm in (4.7).

The conformal background contributes gauge kinetic weight $(1+\eta f_t)$: its four-dimensional volume factor and dilaton factor leave one power of $1+\eta f_t$. The extra inverse internal metric in the gauge mass term cancels that power to first order. Thus the direct gauge correction to the pencil is $T_t^* T_t - \lambda f_t$.

Eliminating both complete leading heavy towers gives

$$E_t(\lambda) = P_t \left[T_t^* T_t - \lambda f_t - 2 J_t^* (\Delta_1 - \lambda)^{-1} J_t \right] P_t. \tag{4.8}$$

Each factor of two in this expression comes from one of the two distinct, equally normalized heavy sectors. No finite KK cutoff is taken. Leading elliptic estimates give $\|p\|_{H^k} \le C_k \|p\|$

for $p \in \operatorname{im} P_t$, uniformly in t. Together with the one-form gap and the uniform Poisson bound, this proves, on a fixed interval below half that gap,

$$\|E_t(\lambda)\| + \|\partial_\lambda E_t(\lambda)\| \le C. \tag{4.9}$$

There is no inverse small gauge eigenvalue in this bound.

For clarity, the order count also covers the terms not visible in (4.7). Metric–two-form heavy mixing from the background H and first-order changes of their operators are of order η. Between two gauge–heavy mixings of order $\sqrt{\eta}$ they first affect the low pencil at order η^2. The $\eta\|I_t v\|^2$ term has the same property. A gauge-vector induced torsion variation is of order $\sqrt{\eta}$ in canonical variables, so its quadratic contribution through $\eta R(\Theta)^2$ starts at order η^2. Quadratic gauge Chern–Simons terms have at least two external legs and cannot contract with the purely internal background to supply an earlier diagonal term. Lorentz Chern–Simons corrections to a gauge–heavy mixing start at order $\eta^{3/2}$. Finally the off-diagonal coupling to the gapped gauge complement starts at order η, and its Schur contribution at order η^2. The absence of a further independent bosonic α'^2 operator in the chosen scheme matters here: canonical gauge normalization removes one power of η from a quadratic gauge term. We extract a perturbative coefficient; we do not treat the raw curvature-squared action as a positive bounded perturbation of an infinite-dimensional second-order form.

4.3 Hodge cancellation, positivity and the error bound

Let H_2 denote harmonic projection on real two-forms. For every adjoint zero-form p, ASD gives $*T_t p = -T_t p$. Decompose $T_t p = h + e + c$ into harmonic, exact and coexact parts. Hodge star preserves h and exchanges e, c isometrically, so $\|e\| = \|c\|$. Moreover $e = d\Delta_1^{-1} d^* T_t p$. Hence

$$\|T_t p\|^2 - 2\langle J_t p, \Delta_1^{-1} J_t p\rangle = \|\mathsf{H}_2 T_t p\|^2. \tag{4.10}$$

This elementary identity is a useful check on the two heavy normalizations. Polarization and the resolvent identity rewrite (4.8) as

$$\begin{aligned} E_t(\lambda) &= P_t T_t^* \mathsf{H}_2 T_t P_t - \lambda Z_t(\lambda), \\ Z_t(\lambda) &= P_t \left[f_t + 2 J_t^* \Delta_1^{-1} (\Delta_1 - \lambda)^{-1} J_t \right] P_t. \end{aligned} \tag{4.11}$$

Define the first-order effective model by

$$F_t^{[1]}(\lambda, \eta) = A_t - \lambda I + \eta E_t(\lambda), \qquad m^{2,[1]} = L^{-2}\lambda. \tag{4.12}$$

For small η, its zero-energy mass form is nonnegative, and its kinetic form $I + \eta Z_t(\lambda)$ is positive on the fixed low interval. Also $\partial_\lambda F_t^{[1]} = -I + O(\eta)$ is negative definite there. These assertions concern a finite 96-dimensional pencil with an exactly specified coefficient.

Every parallel generator of the unbroken algebra has $d_{\mathcal{A}_t} p = 0$ and $T_t p = 0$ pointwise. In the central unitary directions the latter follows from determinant-flat curvature on each

stable factor; it also holds in the trivial SO(14) block. On an axis the separate rank-three determinants remain trivial because $\mathrm{Pic}^0(S) = 0$. Consequently

$$\ker F_t^{[1]}(0,\eta) = \ker A_t. \tag{4.13}$$

There are exactly 93, 94 or 96 zero roots on the corresponding strata. Positivity precludes an extra kernel outside $\ker A_t$.

Proposition 4.1. *For sufficiently small $a, b, \eta \geq 0$, the pencil (4.12) has 96 real low roots, counted with multiplicity, with kernels as in (4.13). Its three positive generic branches satisfy (1.2).*

Proof. Choose a fixed positive cutoff below the one-form and gauge complementary gaps and then shrink the parameter neighborhood so that the whole leading low cluster lies well below it. The pencil is positive at a small fixed negative endpoint and negative at the positive endpoint, and decreases strictly between them. Its ordered eigenvalue functions therefore have exactly one zero each. Equation (4.9) and the min–max inequalities displace each root from the corresponding eigenvalue of A_t by at most $C\eta$. Divide by the positive finite eigenvalue, whose minimum is a fixed constant times b^2, and apply Proposition 3.1. Exact zero branches are treated by (4.13), not by division. □

The formal full expression is $F = A_t - \lambda I + \eta E_t(\lambda) + O_{\mathrm{formal}}(\eta^2)$. Here O_{formal} records uncomputed higher coefficients; it is not a uniform numerical bound on an exact finite-η pole. Solving (4.12) retains its energy dependence and thereby resums some higher powers by convention. Those powers are not additional calculated string coefficients. Along $b = \varepsilon^p$, $a = \varepsilon$, $p > 1$, the sufficient condition $\eta = o(\varepsilon^{2p+2})$ refers precisely to the model estimate just proved.

5 Derivative estimates and a nonzero curvature component

5.1 Preserving the quadratic rate in holomorphic coordinates

To use (2.10) in an interaction coefficient we need two metric derivatives. Its quadratic rate cannot simply be differentiated. The additional input is that both metrics are Ricci-flat Kähler in the same complex structure.

Lemma 5.1. *Let g_* be a smooth Ricci-flat Kähler metric on a complex manifold U, and let $K \Subset U_1 \Subset U$. Suppose g_t are Ricci-flat Kähler for the same complex structure and $\sup_{U_1} |g_t - g_*|_{g_*} \leq e_t \to 0$. Then for every fixed integer $k \geq 0$ and $0 < \alpha < 1$,*

$$\|g_t - g_*\|_{C^{k,\alpha}(K,g_*)} \leq C_{K,U_1,g_*,k,\alpha} e_t \tag{5.1}$$

for all sufficiently small t. The comparison uses the original complex coordinates, without a parameter-dependent coordinate change.

Proof. Cover K by finitely many fixed holomorphic coordinate balls compactly contained in larger balls in U_1. Uniform metric equivalence and [21, Proposition 2.1] give uniform bounds for every fixed coordinate derivative on intermediate balls. These estimates alone do not preserve the rate e_t.

For a Ricci-flat Kähler metric, differentiation of $\log\det g$ and the Kähler interchange of derivatives give

$$g^{p\bar{q}}\partial_p\partial_{\bar{q}}g_{i\bar{j}} = g^{p\bar{q}}g^{r\bar{s}}(\partial_i g_{p\bar{s}})(\partial_{\bar{j}}g_{r\bar{q}}). \tag{5.2}$$

Subtract the equations for g_t and g_*, and write $h = g_t - g_*$. The inverse difference is $g_*^{-1} - g_t^{-1} = g_t^{-1}hg_*^{-1}$. Integrating the differential of the quadratic right side along $g_* + sh$, $0 \le s \le 1$, gives a homogeneous linear system

$$g_t^{p\bar{q}}\partial_p\partial_{\bar{q}}h = A_t \cdot Dh + B_t h. \tag{5.3}$$

D here includes all real first derivatives. The principal symbol is scalar times the identity on Hermitian components. Its ellipticity and all needed coefficient norms are uniform on intermediate balls by the preceding regularity estimate.

For clarity, lower-order coupling does not require a maximum principle for a system. Apply scalar interior Schauder estimates componentwise in the usual boundary-distance-weighted Hölder norms on an intermediate ball. They give

$$|h|^*_{2,\alpha} \le C\|h\|_0 + C_1\big(|h|^*_{1,\alpha} + |h|^*_{0,\alpha}\big).$$

Weighted interpolation bounds the parenthesis by $\delta|h|^*_{2,\alpha} + C_\delta\|h\|_0$. Choose $C_1\delta < 1/2$ and absorb. The weights are bounded below on a smaller ball, yielding the ordinary $C^{2,\alpha}$ bound by $\|h\|_0 \le e_t$. Differentiating (5.3) and using nested balls gives each higher order inductively, since every differentiated forcing term has the already established rate. The finite cover proves (5.1). □

Let δ_ε be cone dilation and put

$$g_\varepsilon = \varepsilon^{-2}\delta_\varepsilon^*\Psi^* g^{\mathrm{KS}}. \tag{5.4}$$

On any compact subset K of the smooth-simple annulus $\mathcal{C} \cap \{1/4 < r < 4\}$, both g_ε and $g_{\mathcal{C}}$ are Ricci-flat Kähler in the same holomorphic chart. Equation (2.10), on a slightly larger regular neighborhood, and Lemma 5.1 give

$$\|g_\varepsilon - g_{\mathcal{C}}\|_{C^{k,\alpha}(K,g_{\mathcal{C}})} \le C_{K,k,\alpha}\varepsilon^2. \tag{5.5}$$

This is a standard elliptic consequence of the imported quadratic comparison, not a new general regularity theorem.

The dependence on the smooth chart can also be made explicit. Call a holomorphic embedding $\varphi : B_{\mathbb{C}^n}(0,2) \to \mathcal{C}_{\mathrm{simple}}$ admissible at scale $0 < \rho \le 1$ if its image lies in $1/4 < r < 4$ and

$$\Lambda^{-1}g_{\mathrm{Eucl}} \le \rho^{-2}\varphi^* g_{\mathcal{C}} \le \Lambda g_{\mathrm{Eucl}} \tag{5.6}$$

for a fixed Λ. Apply the same proof to the two pulled-back metrics divided by ρ^2. The reference derivatives are uniformly bounded by the same local Ricci-flat estimate, so at the chart center

$$\begin{aligned} |\nabla^k(g_\varepsilon - g_{\mathcal{C}})|_{g_{\mathcal{C}}} &\le C_k \varepsilon^2 \rho^{-k}, \\ |\operatorname{Rm}(g_\varepsilon) - \operatorname{Rm}(g_{\mathcal{C}})|_{g_{\mathcal{C}}} &\le C \varepsilon^2 \rho^{-2}. \end{aligned} \tag{5.7}$$

Use the same tensor type on both sides of the curvature difference. No lower bound for this admissible ρ in terms of b/a is asserted. Such a lower bound, not merely the existence of very small charts, would be needed to turn (5.7) into an explicit estimate uniform toward the Pauli axis.

5.2 What fails near a changing stabilizer

There is a direct diagnostic within the balanced locus. Add $X_3 = c\sigma_2$ to (2.11), with a, b, c real and $Y = 0$. On the real two-plane of components $(X_2\sigma_2, X_3\sigma_1)$, the orthogonal projection onto the orbit of $i\sigma_3$ is

$$P(b,c) = \frac{1}{b^2 + c^2} \begin{pmatrix} b^2 & -bc \\ -bc & c^2 \end{pmatrix}. \tag{5.8}$$

The other orbit generators occupy mutually orthogonal component planes. At $c = 0$,

$$\partial_c P = \begin{pmatrix} 0 & -b^{-1} \\ -b^{-1} & 0 \end{pmatrix}, \qquad \partial_c^2 P = \begin{pmatrix} -2b^{-2} & 0 \\ 0 & 2b^{-2} \end{pmatrix}. \tag{5.9}$$

Their operator norms are b^{-1} and $2b^{-2}$. Thus the horizontal projection in the original balanced coordinates is not uniformly C^2 as $b \to 0$. This does not invalidate the relative C^0 tensor comparison, and alone it is not an intrinsic curvature calculation. The following calculation supplies the latter.

5.3 Curvature in additional harmonic deformation directions

At a Pauli point with $a, b > 0$, choose

$$u: \ \delta X_3 = \sigma_1/\sqrt{2}, \qquad v: \ \delta X_3 = \sigma_2/\sqrt{2}, \tag{5.10}$$

with all other components zero. They are orthonormal for (2.6). Their support is orthogonal to the orbit and its three quaternionic images, so they are horizontal tangent vectors to the hyperkähler quotient.

Proposition 5.2. *With curvature sign positive on a round sphere,*

$$K_{g_{\mathcal{C}}}(u, v) = \frac{3}{2b^2}. \tag{5.11}$$

For $a = \varepsilon$, $b = \kappa\varepsilon$ at fixed $0 < \kappa \le 1$, transport this plane by Ψ and normalize using the actual harmonic metric. Its sectional curvature satisfies

$$K_{g^{\mathrm{KS}}} = \frac{3}{2\kappa^2\varepsilon^2} + O_\kappa(1). \tag{5.12}$$

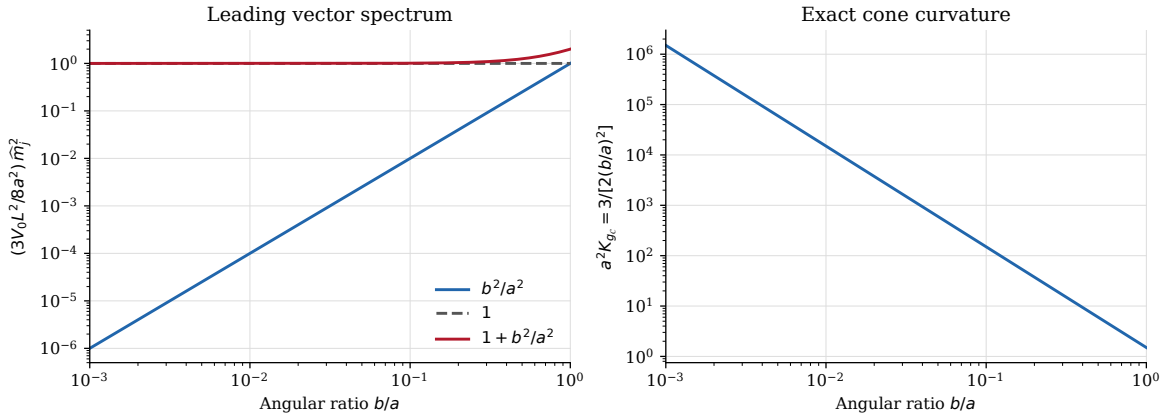


Figure 1. Exact finite-model formulas as the angular ratio decreases. Left: the three normalized leading squared vector masses. The relative geometric estimate remains uniform over the plotted ratios. Right: the cone curvature of the horizontal two-plane (5.10). Its transfer to the actual metric has been proved at each fixed positive ratio; the plot does not assert a ratio-uniform differentiated error. Both panels plot analytic formulas, not numerical HE solutions.

Proof. The effective PU(2) action is free at this point and the moment map is a submersion. Its zero set N is a smooth submanifold of the flat representation. Every tuple of Hermitian X_j with $Y = 0$ lies in N. The affine plane $x + su + tv$ therefore lies in N, and its three relevant second fundamental form entries vanish. The Gauss equation gives $K_N(u, v) = 0$.

Let $D_x\xi = ([\xi, X_j], [\xi, Y_j])$ and use the bilinear orbit Gram form $Q_x(\xi, \zeta) = \langle D_x\xi, D_x\zeta\rangle$, viewed as a map from the Lie algebra to its dual. For horizontal u, v, set $\beta(\xi) = \langle \xi \cdot u, v\rangle$. Differentiating the one-form $\langle D_x\xi, \cdot\rangle$ gives $2\beta(\xi)$. Thus the curvature of the mechanical connection is $2Q_x^{-1}\beta$, up to an irrelevant sign, and the squared norm of O'Neill's tensor is $\beta(Q_x^{-1}\beta)$. In the unnormalized basis $(i\sigma_1, i\sigma_2, i\sigma_3)$,

$$Q_x = \mathrm{diag}(8a^2, 8(a^2 + b^2), 8b^2), \qquad \beta = (0, 0, -2). \tag{5.13}$$

Q_x here is the bilinear Gram matrix, not the matrix of $D_x^*D_x$ after a Hilbert–Schmidt identification of the Lie algebra with its dual. The resulting norm is $1/(2b^2)$. O'Neill's formula [22, 23] gives $K_{N/\,\mathrm{PU}(2)} = K_N + 3\|\mathcal{A}_{uv}\|^2$, proving (5.11).

The ray with fixed κ lies in a fixed regular angular patch; its cone radius is $\sqrt{2(1+\kappa^2)}\varepsilon$. The C^2 estimate (5.5) changes sectional curvature of the rescaled metric by $O_\kappa(\varepsilon^2)$. Scaling back gives (5.12). In particular its relative error is $O_\kappa(\varepsilon^2)$. □

Equation (5.12) is so far a bundle-metric statement. The next section identifies its leading physical meaning without freezing the other neutral fields.

6 A leading neutral-scalar interaction

6.1 Partial reduction on a regular patch

Fix a relatively compact generic patch with $a, b > 0$, and allow the spectator bundle and K3 geometry to vary in small patches. Write $z = (z_E, z_B)$ for bundle coordinates, y for geometric hypermultiplet coordinates, and χ^I for locally unwrapped periodic two-form coordinates. The tensor multiplet and gravity remain explicit. At energies below the smallest Higgs mass we eliminate the three massive vector multiplets as well as the gapped KK modes. We retain all unbroken gauge and charged matter fields. The reduced scalar kinetic metric is a smooth gauged target near this vacuum; it is not the possibly singular vacuum quotient by the unbroken group.

The nondegeneracy needed for this partial reduction can be checked in the local model. Let $D : \mathfrak{su}(2) \to T_x V$ be the broken orbit map. At moment level zero its quaternionic images obey

$$\langle I_\alpha D\xi, I_\beta D\zeta\rangle = \delta_{\alpha\beta} Q_x(\xi, \zeta). \tag{6.1}$$

The symplectic pairings of orbit vectors vanish by moment-map equivariance. Since D is injective, the three moment equations have an invertible derivative on these nine real normal directions. Their norm-square potential has positive Hessian there. The three orbit directions are Goldstone directions; a local gauge slice removes them. This accounts for the twelve real scalar directions in the three massive vector multiplets.

At the actual ASD background the same construction has an elliptic description. If H is its full residual compact gauge group and $\mathfrak{h}$ its parallel algebra, the hyperkähler Weitzenböck formula identifies the ASD cokernel with $\mathfrak{h} \otimes \mathbb{R}^3$. On an H-invariant Coulomb slice the complement of this cokernel has a bounded elliptic right inverse; its finite kernel is the full retained ASD deformation space. For an off-shell effective action solve the heavy Euler–Lagrange equations on this complement. Their leading linearization is the positive ASD Hessian. The implicit-function construction gives a smooth H-equivariant solution in all retained variables, with the unbroken moment maps left as a potential. It agrees with the projected ASD construction on neutral ASD vacua; these equations need not agree away from that locus. The metric, two-form and fermionic complementary operators are those specified in section 3. All inverse bounds here belong to the fixed regular patch, and can deteriorate when its Higgs gap vanishes.

The perturbative expansion below means the finite jet of this tree action, order by order in η and external derivatives. The coefficient functions and the parameter derivatives actually used are smooth on the fixed patch. The notation $O_{\text{formal}}(\eta^j)$ specifies the absence of lower coefficients; it does not bound an unconstructed infinite remainder at finite η.

6.2 Mixed constraints and the vertical kinetic coefficient

For a pure bundle velocity at fixed leading geometry, use a parameter gauge connection Λ_i and set

$$a_i = \partial_i \mathcal{A} - d_{\mathcal{A}} \Lambda_i, \qquad d^*_{\mathcal{A}} a_i = 0, \qquad d^+_{\mathcal{A}} a_i = 0. \tag{6.2}$$

The external-index gauge equation is precisely the first condition; $\mathcal{A}_\mu = \Lambda_i \partial_\mu z^i$ gives $F_{\mu m} = a_{i,m} \partial_\mu z^i$. For stable determinant-fixed E, B, the adjoint Laplacian is invertible off its retained central kernel. Smooth slice dependence and elliptic differentiation control Λ_i and any fixed finite number of derivatives on the patch. K3 geometric variations lift to nearby ASD connections: the trace-free stable ASD cokernel is zero, since its harmonic self-dual adjoint forms have parallel coefficients. The neutral central equations are compatible with the fixed trivial determinants.

Put $f_i = \partial_i f$ and $(j_i)_m = B_F(F_{mn}, a_i{}^n)$. For the convention $G_{\mu m} = \eta v_{i,m} \partial_\mu z^i$ the mixed Einstein equation to the required order is

$$\Delta_1 v_i = df_i + j_i, \qquad d^* v_i = 0. \tag{6.3}$$

Its coefficients follow directly from (4.1): the conformal internal metric contributes $-3\eta\, df_i/2$, the dilaton contributes $+\eta\, df_i$, the mixed metric contributes $\eta d^* dv_i/2$, and the gauge stress contributes $-\eta j_i/2$. The H stress starts at η^2; the mixed R^2 source for a pure bundle velocity does also, because the leading tangent connection is independent of z.

The longitudinal equation is essential. Differentiating (4.5), using the Yang–Mills equation and trace invariance, gives

$$\Delta_0 f_i = \tfrac{1}{2} B_F(F_{mn}, (d_{\mathcal{A}} a_i)^{mn}) = B_F(F_{mn}, D^m a_i{}^n) = -d^* j_i. \tag{6.4}$$

Thus $df_i + j_i$ is co-closed. Since $H^1(S, \mathbb{R}) = 0$, (6.3) has the controlled solution $v_i = \Delta_1^{-1}(df_i + j_i)$.

For the two-form constraint, write $H_S = \eta h_1$ and $H_{\mu mn} = \eta (\mathcal{H}_i)_{mn} \partial_\mu z^i$. The mixed Bianchi identity and the external-index two-form equation give

$$d\mathcal{H}_i = \partial_i h_1 + B_F(a_i \wedge F) =: S_i, \qquad d^* \mathcal{H}_i = 0. \tag{6.5}$$

The source is closed: the derivative of the internal Bianchi identity is $d\partial_i h_1 = -B_F(d_{\mathcal{A}} a_i \wedge F)$, and the second term cancels it using $d_{\mathcal{A}} F = 0$. Since $H^3(S, \mathbb{R}) = 0$, with G_3 the ordinary Green operator,

$$\mathcal{H}_i = d^* G_3 S_i + h_i, \tag{6.6}$$

where h_i is harmonic. These harmonic components describe the retained periodic two-form coordinates and their local connection; they are not inverted. Their parameter-space curvature is the harmonic projection of the universal Chern–Weil form, including the parameter legs of the universal gauge connection. The universal Bianchi identity makes it closed. On a contractible patch a smooth connection potential is obtained by the Poincaré homotopy operator. This accounts for parameter integrability; setting every h_i to zero independently would not do so. All computed coefficients and their finite parameter derivatives are bounded on the patch.

The action now determines the leading bundle tensor, not merely its expected scale. The mixed Yang–Mills term is order η and gives the L^2 metric of (6.2). The leading metric

and dilaton are independent of z; their bundle dependence and metric compensator start at order η, so their pure-bundle kinetic contribution starts at η^2. The two-form constraint has the same order, with its periodic-coordinate connection of order one. The leading tangent connection is independent of z; its variation through the corrected metric and torsion starts at η. The curvature-squared and Lorentz Chern–Simons terms therefore do not add an order-η pure-bundle tensor. They are not omitted from base and higher-order coefficients.

If $a_i = \beta_i - \beta_i^*$ with β_i its $(0,1)$ part, the real one-form norm is twice the Dolbeault norm. Restoring the normalization (3.1) gives

$$S_{\text{scalar}} = -\frac{1}{2}\int C_B g^{\text{KS}}_{ij}\partial_\mu z^i_E \partial^\mu z^j_E + \cdots, \qquad C_B = \frac{2L^2}{g_{10}^2}. \tag{6.7}$$

This is a block of the full kinetic tensor, not yet its intrinsic or ambient curvature.

6.3 The other neutral fields and ambient curvature

After a common six-dimensional normalization, the neutral metric has the following finite-order structure:

$$\begin{aligned} G_\eta &= g_{AB}(y)dy^A dy^B + \eta h_{ij}(y,z)(dz^i + C^i_A dy^A)(dz^j + C^j_B dy^B) \\ &\quad + \eta^2 k_{IJ}(y)(d\chi^I + V^I)(d\chi^J + V^J) + E_\eta, \end{aligned} \tag{6.8}$$

$$h(y,z) = V_0^{-1}\big(g^{\text{KS}}_{E,y}(z_E) \oplus g^{\text{KS}}_{B,y}(z_B)\big). \tag{6.9}$$

The leading blocks g, h, k are positive and smooth. The connection V has both dy and dz components. Perturbative two-form shift symmetry makes the coefficient functions independent of χ. With two parameter derivatives, the formal remainder orders are

$$\begin{gathered} E_{yy} = O_{\text{formal}}(\eta), \quad E_{yz}, E_{zz} = O_{\text{formal}}(\eta^2), \\ E_{y\chi}, E_{z\chi}, E_{\chi\chi} = O_{\text{formal}}(\eta^3). \end{gathered} \tag{6.10}$$

The leading y, z cross term is included in C. The leading two-form Hodge metric depends only on y; its first correction is one order higher. These orders follow from the constraints and action count above and agree with the geometry–bundle–two-form hierarchy of [5]. The Lorentz Chern–Simons and curvature-squared contributions identified there are included at their required orders.

We spell out the curvature consequence because the small axion block cannot be treated as a bounded spectator. In the coframe of (6.8) the scales are $(1, \eta, \eta^2)$. In coordinates the inverse orders are

$$\begin{gathered} G^{yy} = O(1), \quad G^{yz} = O(1), \quad G^{zz} = \eta^{-1}h^{-1} + O(1), \\ G^{z\chi} = O(\eta^{-1}), \quad G^{y\chi} = O(1), \quad G^{\chi\chi} = O(\eta^{-2}). \end{gathered}$$

For lower-index Christoffel symbols with two bundle arguments,

$$\Gamma_{z,zz} = \eta\Gamma(h)_{z,zz} + O(\eta^2), \quad \Gamma_{y,zz} = O(\eta), \quad \Gamma_{\chi,zz} = O(\eta^2).$$

The last estimate uses shift independence. In the coordinate curvature formula the second-derivative terms and the zz Christoffel contraction give $\eta R(h)$; all other contractions have order η^2 or higher, including the one with $G^{\chi\chi}$. Consequently

$$R(G_\eta)_{EEEE} = \frac{\eta}{V_0} R(g_E^{\mathrm{KS}}) + O_{\mathrm{formal}}(\eta^2). \tag{6.11}$$

The leading product in (6.9) is essential for the pure-E coefficient. The spectator-bundle moduli have not been declared constant. A nonsingular tensor-field block can also be retained; its four-E contributions start one order later by the same calculation. In product-frame units the leading sectional curvature is therefore $C_B^{-1} K_{g^{\mathrm{KS}}}$.

It remains to compare neutral curvature with the ambient scalar target including charged fields. Every additional hypermultiplet in table 1 is charged under the residual compact group. The neutral fields are exactly the E, B moduli and the geometric/two-form fields. The equivariant partial reduction constructed above gives a smooth metric on which the full residual group acts by isometries. Its fixed component is totally geodesic: a geodesic with fixed initial point and fixed tangent remains fixed by uniqueness. Thus its four-neutral curvature is the ambient scalar curvature. The residual global gauge action, including shared centers, is retained; no further finite or continuous quotient is taken. This argument does not say that the E-only slice is totally geodesic among the neutral fields; their effects have instead been included in (6.8)–(6.11).

6.4 The interaction and its order of limits

Combining (6.7), (6.11) and Proposition 5.2 proves (1.3). In physical scalar normal coordinates with the convention assigning positive curvature to a sphere, $G_{ij} = \delta_{ij} - \frac{1}{3} R_{ikjl} \phi^k \phi^l + \cdots$. The two-polarization part of the Lorentzian action $-\frac{1}{2} G_{ij} \partial\phi^i \partial\phi^j$ is

$$\mathcal{L}_4 = \frac{\mathcal{K}_{\mathrm{phys}}^{\mathrm{lead}}}{6} \left[(\phi^v)^2 (\partial\phi^u)^2 + (\phi^u)^2 (\partial\phi^v)^2 - 2\phi^u \phi^v \partial\phi^u \cdot \partial\phi^v \right] \tag{6.12}$$

at the calculated large-volume order. It is a nonzero curvature-controlled scalar coefficient in the covariant tree amplitude dictionary [6]. Neutral exchange contributions that appear in other coordinates assemble into this coefficient; universal graviton-exchange poles are separate.

The statement first computes the leading tree large-volume coefficient on a fixed regular patch, then takes $a = \varepsilon$, $b = \kappa\varepsilon$ with fixed $\kappa > 0$. The geometric relative error is $O_\kappa(\varepsilon^2)$. The derivative expansion requires $E_{\mathrm{probe}} \ll \kappa\varepsilon/L$, with its next parameter $E_{\mathrm{probe}}^2/m_{\mathrm{min}}^2$, and perturbative interactions in canonical normalization. Taking the probe energy sufficiently small and the ten-dimensional coupling weak leaves a nonempty regime. Keeping a six-dimensional dilaton fixed as volume grows need not keep that ten-dimensional coupling small, so these limits are specified separately. We have not bounded higher physical coefficients as $\varepsilon \to 0$ or proved a uniform simultaneous finite-η curvature remainder.

7 Discussion

The compactification constructed here has a local region in which one can compare actual HE geometry with two physically identified quantities. The leading vector hierarchy is controlled relatively even when its smallest scale is arbitrarily weaker than the others. The defined first-order tree pencil includes the leading metric and two-form vector towers, and its ASD cancellation preserves the full unbroken kernel. The scalar calculation uses the same family, but requires derivative estimates and a compensated reduction to identify a leading ambient interaction coefficient. Its angular restriction is essential.

The two results have different uniformity. The spectral theorem uses energy-relative reconstruction and survives changing orbit scales without differentiating a horizontal projector. Curvature uses smooth holomorphic regularity on a fixed angular patch. The explicit projector derivatives and nonzero cone curvature show why a uniform derivative assertion needs further information. A quantitative lower bound on the admissible holomorphic chart scale would be one route toward such a result.

The hyperkähler quotient, qualitative Higgs mechanism, heterotic bundle kinetic term, Hodge decomposition and curvature–amplitude relation are established ingredients. The substantial analytic premise is the local relative comparison in [1]; the present contribution is its normalized, coupled use in an actual compactification with a worked interaction and explicit error regimes. The local harmonic argument from [10] supplies the needed coordinates without a global claim for the rank-six moduli space.

Several stronger questions remain. An exact finite-α' background with numerical remainder estimates would be needed to turn the vector pencil bound into an exact corrected-pole bound. A simultaneous interaction theorem would require control of higher physical coefficients and their derivatives near enhancement. Finite-coupling corrections require separate estimates. Nonzero-FI wall crossing, bubbling and quantum amplitudes are also outside the present result. At energies above the smallest Higgs mass, the three massive multiplets must be restored rather than extending the quotient sigma model beyond its window.

References

[1] B.J. Wuebben, *Quantitative local models for Hermitian–Einstein metrics*, 7 November, 2026. 10.5281/zenodo.24029090.

[2] L.B. Anderson, J. Gray, A. Lukas and B. Ovrut, *Stability Walls in Heterotic Theories*, *JHEP* **09** (2009) 026 [0905.1748].

[3] G. Honecker, *Massive U(1)s and Heterotic Five-Branes on K3*, *Nucl. Phys. B* **748** (2006) 126 [hep-th/0602101].

[4] J. Louis, M. Schasny and R. Valandro, *6D Effective Action of Heterotic Compactification on K3 with nontrivial Gauge Bundles*, *JHEP* **04** (2012) 028 [1112.5106].

[5] S. Alexandrov, J. Louis, B. Pioline and R. Valandro, *N=2 Heterotic-Type II duality and bundle moduli*, *JHEP* **08** (2014) 092 [1405.4792].

[6] C. Cheung, A. Helset and J. Parra-Martinez, *Geometric Soft Theorems*, *JHEP* **04** (2022) 011 [2111.03045].

[7] P. Candelas, X. de la Ossa and J. McOrist, *A Metric for Heterotic Moduli*, *Commun. Math. Phys.* **356** (2017) 567 [1605.05256].

[8] J. McOrist, M. Sticka and E.E. Svanes, *The moduli of the universal geometry of heterotic moduli*, 2411.05350.

[9] J. McOrist and Q. Yin, *Heterotic moduli, the double extension and the α'^2 metric*, 2607.15817.

[10] B.J. Wuebben, *Canonical metrics on singular moduli of K3 bundles*, 8 November, 2026. 10.5281/zenodo.24060228.

[11] K. Yoshioka, *Irreducibility of moduli spaces of vector bundles on K3 surfaces*, math/9907001.

[12] S.K. Donaldson, *Anti Self-Dual Yang-Mills Connections Over Complex Algebraic Surfaces and Stable Vector Bundles*, *Proc. Lond. Math. Soc.* **50** (1985) 1.

[13] K. Uhlenbeck and S.T. Yau, *On the existence of hermitian-yang-mills connections in stable vector bundles*, *Commun. Pure Appl. Math.* **39** (1986) S257.

[14] N. Budur and Z. Zhang, *Formality conjecture for K3 surfaces*, *Compos. Math.* **155** (2019) 902 [1803.03974].

[15] N. Buchdahl and G. Schumacher, *An Analytic Application of Geometric Invariant Theory*, *J. Geom. Phys.* **165** (2021) 104237 [2008.04625].

[16] M. Itoh, *Gauge fields and quaternion structure*, *Kodai Math. J.* **11** (1988) 344.

[17] X. Dai, X. Wang and G. Wei, *On the Stability of Riemannian Manifold with Parallel Spinors*, *Invent. Math.* **161** (2005) 151 [math/0311253].

[18] E. Bergshoeff and M. de Roo, *The quartic effective action of the heterotic string and supersymmetry*, *Nucl. Phys. B* **328** (1989) 439.

[19] L. Anguelova, C. Quigley and S. Sethi, *The Leading Quantum Corrections to Stringy Kähler Potentials*, *JHEP* **10** (2010) 065 [1007.4793].

[20] J. Gillard, G. Papadopoulos and D. Tsimpis, *Anomaly, Fluxes and (2,0) Heterotic-String Compactifications*, *JHEP* **06** (2003) 035 [hep-th/0304126].

[21] H.-J. Hein and V. Tosatti, *Higher-order estimates for collapsing Calabi-Yau metrics*, *Camb. J. Math.* **8** (2020) 683 [1803.06697].

[22] B. O'Neill, *The fundamental equations of a submersion.*, *Michigan Math. J.* **13** (1966) 459.

[23] R. Bielawski, *Curvature of hyperkähler quotients*, *Central Eur. J. Math.* **6** (2008) 191 [math/0702889].